The short answer is that the f-DP and (ε, δ)-DP curves always contain exactly the same information, and they both always imply the full exact Rényi DP (RDP) curve. However, the reverse implication—that the full exact RDP curve implies the f-DP curve—**only holds under a specific regularity condition** regarding the moment generating function of the privacy loss. If the mechanism has heavy tails such that the RDP curve is strictly infinite for all $\alpha > 1$, the RDP curve fails to uniquely identify the f-DP curve.

1 Setup and Definitions

Let $P = \mathcal{L}(M(x))$ and $Q = \mathcal{L}(M(x'))$ be the distributions of the mechanism on neighboring datasets. Since the additive noise $Z \sim \nu$ is centered and symmetric, P and Q are symmetric shifts of each other.

Define the **privacy loss random variable** under P as:

$$L = \log \frac{dP}{dQ}$$

The three exact privacy curves in question are defined as follows:

1. **The RDP Curve**: For $\alpha > 1$,
$$D_\alpha(P\|Q) = \frac{1}{\alpha - 1} \log \mathbb{E}_P \left[e^{(\alpha-1)L} \right]$$
2. **The (ε, δ)-DP Curve**: For $\varepsilon > 0$,
$$\delta(\varepsilon) = \sup_S [P(S) - e^\varepsilon Q(S)]$$
3. **The f-DP Curve**: The trade-off function $f_{P,Q} : [0, 1] \to [0, 1]$ represents the optimal hypothesis testing error:
$$f_{P,Q}(x) = \inf_\phi \{\mathbb{E}_Q[1 - \phi] : \mathbb{E}_P[\phi] \leq x\}$$

2 Equivalence of f-DP and (ε, δ)-DP

Theorem 1. *The full exact f-DP curve and the full exact (ε, δ)-DP curve uniquely determine each other.*

Proof. By the results of Dong, Roth, and Su (2021), the trade-off function $f_{P,Q}(x)$ is a convex, decreasing function. The relation between f-DP and (ε, δ)-DP is governed by convex duality.

Given the trade-off function $f_{P,Q}(x)$, the optimal $\delta(\varepsilon)$ for any $\varepsilon \geq 0$ is given by:

$$\delta(\varepsilon) = \sup_{x \in [0,1]} [1 - xe^\varepsilon - f_{P,Q}(x)]$$

Conversely, because $f_{P,Q}$ is convex and bounded below by 0, it is equal to its biconjugate. We can uniquely recover $f_{P,Q}(x)$ from the (ε, δ)-DP curve via:

$$f_{P,Q}(x) = \sup_{\varepsilon \geq 0} [1 - xe^\varepsilon - \delta(\varepsilon)] \quad \text{for } x \in [0, 1]$$

Thus, $\{f(\alpha) : 0 \leq \alpha \leq 1\}$ and $\{\delta(\varepsilon) : \varepsilon > 0\}$ contain identical information. □

3 The f-DP Curve Implies the RDP Curve

Theorem 2. *The full exact f-DP curve uniquely determines the exact RDP curve $\{D_\alpha(P\|Q) : \alpha > 1\}$.*

Proof. By the Neyman-Pearson lemma, the optimal test ϕ for testing P vs Q at level x consists of thresholding the likelihood ratio $dP/dQ = e^L$. Therefore, the trade-off function $f_{P,Q}$ is exactly the flipped Receiver Operating Characteristic (ROC) curve of the likelihood ratio test.

The distribution of the privacy loss L is fully encoded in the geometry of $f_{P,Q}$. Specifically, because $f_{P,Q}$ is convex, it has a well-defined right-derivative $f'_{P,Q}(x)$ almost everywhere. The slopes of $f_{P,Q}$ correspond to the negative thresholds of the likelihood ratio:

$$-f'_{P,Q}(x) = c = e^\ell$$

Consequently, the full trade-off curve $f_{P,Q}$ uniquely identifies the cumulative distribution function (CDF) of the likelihood ratio under both P and Q. Because f-DP uniquely identifies the probability law $\mathcal{L}(L)$, it uniquely determines all moments of e^L.

Since $D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log \mathbb{E}_P[e^{(\alpha-1)L}]$, the f-DP curve directly evaluates the RDP curve for all $\alpha > 1$. □

4 Does the RDP Curve Imply the f-DP Curve?

This requires a vital regularity condition on the tails of the noise distribution ν.

Assumption (Local Finiteness): *There exists some $\alpha_0 > 1$ such that $D_\alpha(P\|Q) < \infty$ for all $\alpha \in (1, \alpha_0)$.*

Theorem 3. *Under the Local Finiteness assumption, the exact RDP curve uniquely determines the f-DP curve. If this assumption fails (i.e., $D_\alpha = \infty$ for all $\alpha > 1$), the RDP curve does* not *uniquely determine the f-DP curve.*

Proof (When the condition holds). Let $M_L(t) = \mathbb{E}_P[e^{tL}]$ be the moment generating function (MGF) of L. For $t > 0$, we have $M_L(t) = \exp(t \cdot D_{t+1}(P\|Q))$.

Let us examine $M_L(t)$ for $t \in (-1, 0]$. Using the change of measure, we have:

$$M_L(t) = \mathbb{E}_P\left[\left(\frac{dP}{dQ}\right)^t\right] = \mathbb{E}_Q\left[\left(\frac{dP}{dQ}\right)^{t+1}\right]$$

Since $t \in (-1, 0]$, we have $t + 1 \in (0, 1]$. Because the function $z \mapsto z^{t+1}$ is concave, Jensen's inequality implies $\mathbb{E}_Q[(dP/dQ)^{t+1}] \le (\mathbb{E}_Q[dP/dQ])^{t+1} = 1$. Thus, $M_L(t) \le 1$ for all $t \in (-1, 0]$.

By the Local Finiteness assumption, $M_L(t) < \infty$ for $t \in (0, \alpha_0 - 1)$. Combining these, $M_L(t)$ is finite on the open interval $(-1, \alpha_0 - 1)$.

A standard theorem in probability (e.g., properties of the bilateral Laplace transform) states that if the MGF of a random variable is finite on an open interval containing the origin, it uniquely determines the probability distribution. Because 0 is strictly inside $(-1, \alpha_0 - 1)$, the exact RDP curve uniquely determines $\mathcal{L}(L)$.

As established in Theorem 2, the probability law of the likelihood ratio uniquely determines the Neyman-Pearson trade-off function $f_{P,Q}(x)$. Therefore, the RDP curve implies the f-DP (and (ε, δ)-DP) curves. □

Proof of the Converse Failure (When the condition is violated). Suppose we choose an additive noise mechanism $Z \sim \nu$ with extremely heavy tails such that the likelihood ratio has no bounded moments strictly greater than 1 (for instance, certain parameterizations of the Cauchy distribution or Log-Cauchy).

In this case, $D_\alpha(P\|Q) = \infty$ for all $\alpha > 1$. The exact RDP curve is simply the constant infinity. However, there are infinitely many distinct heavy-tailed distributions that yield $D_\alpha = \infty \quad \forall \alpha > 1$, and these distributions will have strictly different optimal hypothesis testing bounds (different f-DP curves). Therefore, knowing the RDP curve is ∞ provides zero information to reconstruct the specific f-DP curve. □